

Chapter 8 – Functions with Arrays and Strings

Slide 1 — What Are Functions with Arrays and Strings?

Definition

Functions can work with **arrays and strings** by receiving them as arguments.

An **array** stores multiple values of the same data type.

A **string** is a sequence of characters stored in a character array.

Instead of writing the same operation repeatedly, we can send an array or string to a function and let the function process it.

Analogy

Think of a delivery box:

- **Function** → Worker
- **Array** → Box containing many items
- **String** → Box containing characters
- **Function call** → Giving the box to the worker
- **Result** → Processed information

Small Example

None

```
void display(int arr[], int n) {  
    printf("%d", arr[0]);  
}
```

Here, `arr[]` allows the function to work with an array.

Remember

A function can receive an entire array or string instead of handling every value separately.

Slide 2 — Why Are Arrays and Strings Passed to Functions?

Why do we use them?

Passing arrays and strings to functions helps us:

- Process multiple values using one function.
- Avoid repeating the same code.
- Search or calculate values inside an array.
- Display array elements.
- Process text using a string.
- Make programs more organized and reusable.

Basic → Professional

Basic:

```
None  
void display(int arr[], int n)
```

The function receives an array and its size.

Professional idea:

A program can use separate functions for tasks such as:

```
None  
Input Data  
  ↓  
Process Array  
  ↓  
Search / Calculate  
  ↓
```

Display Result

Important Statement

Functions make array and string operations reusable instead of writing the same logic again and again.

Key Point

When an array is passed to a function, the function can work with its elements.

Slide 3 — Arrays vs Strings in Functions

Arrays

An array can contain multiple values:

None

```
int marks[] = {80, 75, 90};
```

A function can receive the array:

None

```
void show(int marks[], int n)
```

Here:

- `marks[]` → Array
- `n` → Number of elements

Strings

A string is a character array:

None

```
char name[] = "Arun";
```

It can be passed to a function:

None

```
void display(char name[])
```

Comparison

Array	String
Stores multiple values	Stores characters/text
Can contain <code>int</code> , <code>float</code> , etc.	Uses <code>char</code>
Usually passed with its size	Ends with <code>'\0'</code>
Used for collections of data	Used for text

Important Statement

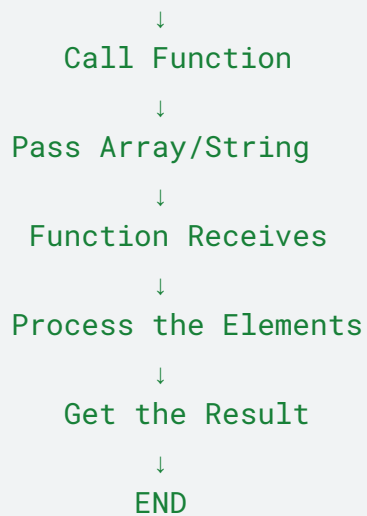
A string in C is a character array ending with the null character `'\0'`.

Slide 4 — Flowchart: Function with Array/String

Flow

None

```
graph TD
    START[START] --> Create[Create Array/String]
```



Example

None

```
#include <stdio.h>

void display(int arr[], int n) {
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
}

int main() {
    int numbers[] = {10, 20, 30};

    display(numbers, 3);

    return 0;
}
```

Output

None

```
10 20 30
```

What Happened?

1. `numbers` contains three values.
2. The array is passed to `display()`.
3. The function receives the array.
4. Each element is processed.
5. The values are displayed.

Remember

Create → Pass → Process → Display

Slide 5 — Witty Recap

Imagine a Delivery Team

You have a box full of items.

- **Array** → Box containing many items
- **String** → Box containing letters
- **Function** → Worker processing the box
- **Argument** → The box you hand over
- **Function execution** → Worker processes the contents
- **Output** → Final result

Practice Questions

1. What can a function receive to work with multiple array elements?

- A. Only one integer
- B. An array
- C. Only a character
- D. Only a return value

 **Hint:** Think about sending a collection of values to a function.

 Glossary:

Array — A collection of elements of the same data type.

Answer: B. An array

2.

None

```
#include <stdio.h>

void display(int arr[]) {
    printf("%d", arr[1]);
}

int main() {
    int numbers[] = {10, 20, 30};

    display(numbers);

    return 0;
}
```

What will be the output?

 **Hint:** Array indexing starts from 0.

 Glossary:

Index — The position used to access an element in an array.

Answer:

None

20

3.

None

```
#include <stdio.h>

void display(_____ name) {
    printf("%s", name);
}

int main() {
    char name[] = "Arun";
    display(name);

    return 0;
}
```

Fill in the missing code.

 **Hint:** The function is receiving a string.

 **Glossary:**

String — A sequence of characters stored in a character array.

Answer: `char *`

4. Arrange the words correctly:

variable / inside / A / function / local / is / declared



Hint: Start with **A**.



Glossary:

Local — A variable whose scope is limited to a particular function or block.

Answer: A local variable is declared inside a function.

5. When a function receives an array and its size, the array and size are passed as _____.



Hint: They are the actual values given during a function call.



Glossary:

Arguments — The actual values passed to a function.

Answer: arguments

Do you like this personality?